Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative disturbance (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (93 sites, 395 site-years) · deadtrees v2-2

ΔRMSE = RMSE(with D) − RMSE(without D)

	XGBoost (Optuna) · 12m	XGBoost (Optuna) · 24m	LightGBM (Optuna) · 12m	LightGBM (Optuna) · 24m
GPPsat (µmol m⁻² s⁻¹)	68% improved	68% improved	68% improved	65% improved
NEPmax (µmol m⁻² s⁻¹)	74% improved	58% improved	76% improved	71% improved
ETmax (mm d⁻¹)	46% improved	41% improved	46% improved	44% improved
WUE (g C mm⁻¹)	57% improved	56% improved	57% improved	58% improved

Relative disturbance (%)
site maximum, 500 m
breaks 21.6 | 34.5

60
40
20
0

Sites (sorted by ΔRMSE within each panel)